Final Test Coverage Report

DDP-087 • Coverage evidence and verification assessment

Darshana Opticals — Optical Shop Management System (OSMS)

DDP-087 | Historical local evidence: 25 October 2026 | Updated 26 October 2026 | Status: evidence-qualified review candidate; not signed off

Document control: Prepared from locally executed Jest/Vitest logs and DDP-087 issue requirements. No manual or E2E outcomes are invented.

1. Executive summary

Two successful local automated test executions were independently recorded: 2,075/2,075 backend tests and 1,341/1,341 frontend tests, totaling 3,416 passing tests across 228 test files/suites. Both measured repositories exceed 80% in each overall code-coverage metric. This is evidence of automated suite health at the tested local revisions, not proof that every approved SRS requirement, API endpoint, release commit, or hosted workflow has been fully verified.

Measure	Backend (Jest)	Frontend (Vitest)
Passing tests	2,075 / 2,075	1,341 / 1,341
Passing suites/files	118 / 118	110 / 110
Failed tests	0	0
Statements	92.34%	88.05%
Branches	81.68%	81.80%
Functions	97.86%	90.95%
Lines	93.00%	88.86%

2. Scope, baseline and evidence provenance

In scope: local Jest backend unit/integration tests, local Vitest/React Testing Library frontend unit/component tests, associated instrumentation and measured code coverage, plus a traceability framework for integration, browser E2E, security, performance, and requirements testing. Out of scope for these two execution logs: independently validated production readiness, hosted payment settlement, exhaustive manual UAT, and full API-to-test or SRS-to-test mapping.

Item	Backend	Frontend
Repository	darshana-opticals-osms/backend	darshana-opticals-osms/frontend
Local tested SHA (previously collected)	47eb042 (reconfirm with git rev-parse HEAD)	3e6fa9b (reconfirm with git rev-parse HEAD)
Local branch (previously collected)	docs/91-post-deployment-security-verification	test/91-owasp-security-verification
Framework	Jest; Supertest where used	Vitest 5.0.5; React Testing Library
Test execution date	25 October 2026	25 October 2026
Recorded duration	265.944 seconds	78.23 seconds

Item	Backend	Frontend
Result artifact	backend/coverage/coverage-summary.json	frontend/coverage/coverage-summary.json

Historical local branch inspections identified backend SHA candidate 47eb042 and frontend SHA candidate 3e6fa9b. Neither tested SHA is embedded in the terminal coverage logs. Therefore, exact execution revision and delivered release parity cannot be certified from current evidence; do not promote these candidates to verified tested SHAs.

3. Reproduction and environment

Backend command: npx jest --runInBand --testTimeout=30000 --coverage --coverageReporters=json-summary --coverageReporters=text-summary. MongoDB Memory Server used the installed MongoDB 8.3.11 executable through session-only MONGOMS_SYSTEM_BINARY; SYSTEM_BINARY_VERSION_CHECK=false and RUNTIME_DOWNLOAD=false. A separate 3/3 database foundation test passed before the full run. The earlier failed run was affected by a stuck MongoDB 8.2.6 binary download; it must not be counted as code regression evidence.

Frontend command: npm run test:coverage -- --coverage.reporter=text --coverage.reporter=json-summary. V8 was the reported coverage provider. Frontend console emitted non-failing React Router future-flag warnings and React act(...) warnings. Record these as maintenance/testing hygiene observations.

4. Coverage target assessment

Guideline requirement	Evidence	Assessment
80% code coverage for new code	All four measured overall metrics >80% in each repository	Overall threshold exceeded; new-code-only coverage not separately measured
90% branch coverage for critical business logic	Overall branch coverage: backend 81.68%; frontend 81.80%	NOT VERIFIED by aggregate numbers; per-critical-module review needed
At least one integration test per API endpoint	Integration tests present; full route inventory/test mapping unavailable	NOT VERIFIED
Critical end-to-end workflows	DDP-084 provides separate execution evidence	Must cross-check actual scenarios, URLs and statuses
Final SRS requirement traceability	Approved SRS requirement inventory not included in execution logs	NOT VERIFIED

5. Backend automated test coverage

Jest reported 118 passed suites and 2,075 passed tests. Overall statement coverage: 6,800/7,364; branch coverage: 4,958/6,070; function coverage: 778/795; line coverage: 6,571/7,065. The additional JSON field branchesTrue=100% is not equivalent to ordinary branch coverage and is not substituted for 81.68%.

Representative observed passing backend test suites include: database foundation; health; admin staff management; staff/doctor lifecycle security; chatbot session API; inventory image lifecycle/upload; loyalty read API; password reset service; prescription service; payment validation/model/monitoring; audit services; reports; authorization middleware; orders, checkout and inventory services. The list establishes observed suite-level execution, not endpoint-by-endpoint integration coverage.

6. Frontend automated test coverage

Vitest reported 110 passing test files, 1,341 passing tests, no failures. Overall coverage is 88.05% statements, 81.80% branches, 90.95% functions and 88.86% lines.

Representative executed areas: sign-up, login/password recovery, protected routes and role navigation; customer profile; product search and detail; cart/checkout/payment entry and result; order history and staff order controls; appointments/doctor sessions; prescriptions; inventory/reorder alerts; loyalty; chatbot; sales/inventory reporting; administrative staff/doctor management. Most frontend tests mock API behavior and should not be conflated with full-stack E2E validation.

7. Module-level risk observations

Frontend module	Observed branch %	Recommended interpretation
Appointment booking page	71.55%	Add boundary/error-state tests if considered critical
Payment entry page	68.70%	Critical payment initiation paths need branch review
Checkout entry page	68.05%	Checkout handling warrants targeted test review
Admin staff editor	66.37%	Sensitive role/reassignment flows warrant review
Staff move appointment page	57.64%	Uncovered move/conflict paths warrant review
Protected routes	100.00%	Strong module result; does not replace server authorization
Payment return page	100.00%	Tested component result only; hosted gateway not proven

These are file-level branch percentages from the frontend console report. The designation “critical” must be finalized by the team against SRS risk categories; no 90% compliance claim is made for them. Backend critical-file percentages require its detailed JSON file, not only the aggregate summary.

8. API integration coverage assessment

A 73-route method/path register is included in the Test Case Register. Candidate suite names are investigation leads rather than proofs of route-level integration. Per-route Supertest HTTP request/assertion and run-result mapping has not been independently confirmed. Accordingly, a 73/73 integration coverage claim is NOT made; uncovered/unknown verification is explicitly recorded.

Endpoint group	Observed test evidence	Full endpoint inventory
Authentication/profile/recovery	Jest authentication and password-reset; frontend auth flows	Pending route-by-route review
Inventory/orders/payments	Jest inventory/checkout/payment; frontend order and payment tests	Pending route-by-route review
Appointments/prescriptions	Jest appointment/prescription; frontend page flows	Pending route-by-route review
Reports/loyalty/chatbot	Jest reporting/loyalty/chatbot; frontend services and flows	Pending route-by-route review
Admin/audit/notifications	Jest admin/audit/notification; frontend	Pending route-by-route review

Endpoint group	Observed test evidence	Full endpoint inventory
	admin flows	

9. E2E, security, performance and defect evidence

DDP-084 (infra issue #11 / merged PR #19) is the source for browser-level integration and end-to-end evidence. Prior project review referenced 10 individually reported Playwright checks, including 7/7 later repeated authorization checks; retain the actual logs and exact environment before promoting these to a final verified matrix. Hosted PayHere callback/payment verification was reported as constrained by a merchant-configuration error; do not label a complete hosted purchase journey “Pass” without new proof.

DDP-085 security: reference the existing documents/security/owasp-checklist.pdf and backend issue #91, including the precise tested conditions, scope and any outstanding limitations. DDP-086 performance: reference documents/testing/performance-report.pdf and backend issue #92; workload, p95 latency, error rate and configured test environment must be copied from that measured report rather than inferred. DDP-092 defect closure: reference infra issue #14 and associated regression evidence; merged status alone is not equivalent to verified zero residual defects.

10. CI and release traceability

No CI workflow run URLs or hosted artifact IDs were embedded in the historical local runner outputs. The previous local candidate SHAs (backend 47eb042, frontend 3e6fa9b) remain unconfirmed for exact coverage execution. The reviewer must compare git rev-parse HEAD from execution-time repositories and GitHub Actions run evidence to delivered release SHAs; absence of such proof is an explicit traceability limitation.

11. Open gaps and corrective follow-up

- Confirm actual tested SHAs, target branch and release commit; attach persistent log/coverage artifacts.
- Map approved SRS v1.0 FR-001–FR-014 and NFR-001–NFR-007 using conservative representative case IDs; track v1.1 deviations against accepted ADRs. Do not treat a passing component test as full requirement acceptance.
- Provide a confirmed HTTP method/path/test assertion for each API route, or retain “endpoint integration unverified” in the 73-route register. Do not substitute unit tests or mocked UI tests.
- Assess each team-designated critical business-logic module against its 90% branch target.
- Cross-check DDP-084 browser E2E and hosted payment outcome with logs; mark unresolved items Blocked/Not Verified.
- Import actual DDP-085, DDP-086 and DDP-092 measured results and traceable issues/CI evidence.

12. Conclusion and sign-off status

Conclusion: historical local Jest/Vitest outcomes passed 3,416/3,416 with all aggregate coverage metrics above 80%. SRS mapping is reconciled at the representative assertion level. Nonetheless, endpoint-level integration proof, 90% branch coverage for all designated critical logic, exact tested and released SHAs, CI artifacts, and complete E2E/hosted PayHere evidence are not yet independently established. Reviewer approval remains pending.

Prepared for DDP-087 review. QA reviewer: _____ Review date: _____ Release SHA(s): _____

Appendix A — SRS requirement alignment

SRS	Business capability	Representative test evidence	Boundary / qualification
FR-001	Authentication and user roles	Jest auth; frontend Login/Signup, route guards	Automated evidence; hosted full-scope not established
FR-002	Customer profile / prescription history	ProfilePage; PrescriptionHistoryPage	Automated evidence
FR-003	Catalogue	ProductsPage; productService	Partial implementation: style filter gap
FR-004	Cart / checkout	CartPage; CheckoutEntryPage; checkout.service	Automated evidence
FR-005	Inventory	InventoryPage; inventory integration tests	Partial implementation: no verified real-time sync
FR-006	Reorder alerts	InventoryReorderAlerts; reorderAlert.service	Automated evidence
FR-007	Doctor sessions / appointment	AppointmentBookingPage; appointment.service	Modified by ADR-007
FR-008	Appointment communication	ADR-009	Approved manual process; automated reminders not in scope
FR-009	PayHere	PaymentEntryPage; payment unit tests	Hosted payment blocked/not verified
FR-010	Loyalty	loyaltyReadApi; CustomerLoyaltyPage	Accrual tested; redemption out of scope
FR-011	Chatbot	chatbotSessionApi; ChatbotFlow	Automated evidence
FR-012	Reports	SalesReportPage; InventoryReportPage	Automated evidence
FR-013	Prescriptions	prescription.service; PrescriptionFlows	Automated evidence
FR-014	Order statuses	OrderDetailsPage; StaffOrderStatusControl	Modified by ADR-004; no courier sync

Appendix B — Source-reviewed API route inventory

The backend source inventory identifies 73 route declarations across 17 route groups. The detailed method, path, access context and possible related test evidence are included in the companion Test Case Register Appendix C. This is an assessment of implementation route declarations only. At least one executed integration assertion per endpoint has NOT been demonstrated for all routes; do not label endpoint coverage complete.

Backend area	Routes inventoried	Endpoint assertion status
auth	4	Not independently verified per endpoint
admin	10	Not independently verified per endpoint
profile	2	Not independently verified per endpoint
product	2	Not independently verified per endpoint
inventory	10	Not independently verified per endpoint
prescription	3	Not independently verified per endpoint
branch	1	Not independently verified per endpoint

cart	4	Not independently verified per endpoint
checkout	1	Not independently verified per endpoint
order	5	Not independently verified per endpoint
payment	9	Not independently verified per endpoint
appointment	10	Not independently verified per endpoint
doctor	2	Not independently verified per endpoint
audit	1	Not independently verified per endpoint
loyalty	2	Not independently verified per endpoint
report	6	Not independently verified per endpoint
health	1	Not independently verified per endpoint

Appendix C — Critical business logic branch assessment

Historical aggregate branch coverage: backend 81.68%; frontend 81.80%. Source log provides frontend file-level data, but no backend per-file coverage table. A 90% branch threshold for critical business modules therefore cannot be certified as achieved from these records.

Selected risk-sensitive module	Measured branch coverage	Assessment
Backend checkout/order/payment/inventory/appointment/authorization	Backend per-file branch metrics unavailable in uploaded text summary	Not verified; export per-file JSON for critical modules
Frontend PaymentEntryPage.jsx	68.70%	Below 90%
Frontend CheckoutEntryPage.jsx	68.05%	Below 90%
Frontend AppointmentBookingPage.jsx	71.55%	Below 90%
Frontend AdminStaffEditor.jsx	66.37%	Below 90%
Frontend CartProvider.jsx	68.08%	Below 90%
Frontend StaffMoveAppointmentPage.jsx	57.64%	Below 90%
Frontend InventoryEditor.jsx	92.40%	At/above 90% for this file
Frontend orderService.js	100%	At/above 90% for this file
Frontend paymentService.js	86.20%	Below 90%

Module criticality requires confirmation by the team. Low coverage above is not automatically a defect but is evidence that full 90% per-critical-module compliance cannot be claimed. Follow-up should add targeted tests to uncovered branches; no production refactoring is requested by this documentation review.

Appendix D — QA execution, CI, and provenance

Backend historical local outcome: 118 of 118 suites, 2,075 of 2,075 tests passing; Jest coverage run duration 265.944 seconds. Frontend historical local outcome: 110 of 110 files, 1,341 of 1,341 tests passing; Vitest 5.0.5 duration 78.23 seconds. Both runs captured on 25 October 2026. Earlier local branch observations: backend 47eb042, frontend 3e6fa9b; uploaded runner output does not prove these SHA values, and they MUST be rechecked before asserting exact baseline.

Local commands: backend npx jest --runInBand --testTimeout=30000 --coverage --coverageReporters=json-summary --coverageReporters=text-summary; frontend npm run test:coverage -- --coverage.reporter=text --coverage.reporter=json-summary. Backend candidate SHA 47eb042; frontend candidate SHA 3e6fa9b; neither SHA

verified by an execution-time git record. CI URLs/artifact IDs unavailable in supplied local logs; do not invent links. These executions are historical local baselines rather than final release sign-off.

Appendix E — Linked integration, security, performance and defect work

Issue	Evidence lead	Outcome qualification
DDP-084	infra#11; infra PR #19	PARTIAL — exact per-scenario browser logs and environment must be checked; hosted PayHere BLOCKED / NOT VERIFIED
DDP-085	backend#91; documents/security/owasp-checklist.pdf	REFERENCED — detailed control findings require independently reviewed checklist
DDP-086	backend#92; documents/testing/performance-report.pdf	PARTIAL — not complete hosted search-to-browser response proof or 99.9% uptime
DDP-092	infra#14	REFERENCED — no claim of zero open defects without final defect register

Hosted PayHere purchase/payment completion stays BLOCKED / NOT VERIFIED. Formal staff usability and hosted 99.9% uptime are also not verified. NFR-001 p95 backend threshold evidence reported for 13 selected local endpoints does not independently prove the SRS browser-display criterion. Any remaining unverified item must be tracked transparently rather than marked PASS.

Appendix F — QA sign-off judgment

SUPPORTED: local automated suite pass status and aggregate code coverage at historical observations. NOT SUPPORTED YET: a complete API endpoint integration pass rate, release-SHA parity, complete CI artifact trace, every critical business module at 90% branches, and hosted PayHere processing. Final QA sign-off remains CONDITIONAL/PENDING until those checks are evidenced. This document includes available assessments and clearly identified residual verification tasks, rather than invented outcomes.

Appendix G — Second reviewer correction cross-check

SRS consistency: authoritative original requirement IDs are FR-001–FR-014 and NFR-001–NFR-007. See corrected Test Case Register Appendix A for exact representative cases; infrastructure-only checks are classified as technical evidence, and SRS v1.1 remains a review candidate.

API coverage: 73 source route declarations have been catalogued, but route-specific passing HTTP integration assertions are not independently demonstrated by the evidence at hand. Confirmation status is “unverified”; do not conflate service-unit or frontend-component results with API integration.

CI provenance: historical local test pass totals and command strings remain measured; exact tested SHAs are only earlier inspected candidate values. No CI run URL, execution-time commit snapshot, permanent logs, or release-SHA equivalence can be confirmed. These are outstanding evidence actions, not document formatting mistakes.

Critical branch thresholds: the recorded low-percentage critical UI modules remain below 90% and the backend critical module detail is absent; no statement of critical target compliance is made.

Cross-repository outcomes: DDP-084 / 085 / 086 / 092 links indicate where supporting evidence must be checked. PayHere hosted callback processing is blocked/not verified in this report until recorded successful execution exists. No application changes requested.

Index: the existing documents README in PR #54 already links coverage-sprint8.pdf, test-register.pdf, performance-report.pdf, and the OWASP checklist. No further README change is required for this correction.

Appendix H — Follow-up evidence audit (26 October 2026)

Confirmed Supertest HTTP request references were independently found in integration test source for 8 of the 73 documented route declarations, including login, inventory image upload, customer/manager loyalty reads, doctor session creation/listing, appointment listing, and staff directory listing. Source links and exact route entries are recorded in the companion Test Case Register Appendix H. Other routes remain unverified by this sample, not necessarily untested.

Workflow lookup for the previously suggested candidate commit SHAs 47eb042 (backend) and 3e6fa9b (frontend) returned no PR-triggered workflow runs. This is a limitation of available evidence and query scope; it is not evidence that CI was never executed. The 25 October test logs prove historical suite totals but not exact execution-time SHAs or release parity.

The 90% critical-business-logic branch target remains not demonstrated: several previously measured critical UI files remain below 90%, and backend critical-module measurements still require retained per-file coverage artifacts. No test should be marked PASSED without its actual execution record. Hosted PayHere remains BLOCKED / NOT VERIFIED.

To achieve final evidence-based sign-off, the team must capture (1) git rev-parse HEAD and timestamp in the same execution session as Jest/Vitest; (2) retained json-summary/HTML or lcov artifacts per critical module; (3) CI workflow URLs or explicitly record CI unavailable; (4) scenario-level DDP-084 and regression evidence, including hosted PayHere outcome when feasible. Documentation alone cannot create missing execution proof.

Appendix I — Fresh local QA execution evidence (26 October 2026)

Evidence basis: newly supplied PowerShell test-console logs and separately observed local Git revisions. The test log itself does not include the Git SHA; evidence files created immediately before the runs should be retained alongside these logs to bind the execution to the revision. No GitHub Actions CI result is implied.

Metric	Backend (Jest)	Frontend (Vitest)	Evidence status	Interpretation
Suite outcome	118/118 passed	110/110 passed	PASS — local	No failing suites/files
Test outcome	2,075/2,075 passed	1,341/1,341 passed	PASS — local	3,416 passing tests combined
Statements	92.34%	88.05%	Measured	Overall statements
Branches	81.68%	81.80%	Measured	Not 90% critical-module proof
Functions	97.86%	90.95%	Measured	Overall functions
Lines	93.00%	88.86%	Measured	Overall lines
Duration	346.634 seconds	77.81 seconds	Measured	Fresh supplied console logs

Execution and source traceability

Backend observed local revision: 47eb042a7627d4ae1f1f67b9c48c370379faa7c5; branch docs/91-post-deployment-security-verification; clean working tree reported before the run. Frontend observed local revision: 3e6fa9b5b277d9663982cb6e36a5af8c8187c97c; branch test/91-owasp-security-verification; clean working tree reported before the run.

Backend command: npx jest --runInBand --testTimeout=30000 --coverage --coverageReporters=json-summary --coverageReporters=text-summary. Test environment configured to use locally installed MongoDB 8.3 mongod with MONGOMS_SYSTEM_BINARY, MONGOMS_SYSTEM_BINARY_VERSION_CHECK=false, MONGOMS_RUNTIME_DOWNLOAD=false.

Frontend command: npm run test:coverage -- --coverage.reporter=text --coverage.reporter=json-summary. Vitest v5.0.5, v8 coverage. Frontend log reports Start at 01:07:09 (local clock; full dated timestamp should be taken from ddp087-frontend-evidence.txt).

Preserve backend ddp087-backend-evidence.txt and ddp087-backend-test.log; frontend ddp087-frontend-evidence.txt and ddp087-frontend-test.log; respective coverage/coverage-summary.json. The evidence text files were created before the test processes, but their actual contents were not supplied here and hence execution-time SHA attestation is pending inspection.

No matching PR-triggered workflow runs were returned for candidate commit lookups; that is not proof that all CI runs are absent. No verified CI URLs or coverage artifact download identifiers are available from the supplied data. The documentation must distinguish local verification from CI and deployed release acceptance.

Representative passing backend HTTP integration suites

The fresh backend log explicitly reports PASS for integration files including paymentApi.test.js, checkoutApi.test.js, orderStatusApi.test.js, appointmentDoctor.api.test.js, prescriptionApi.test.js, cartApi.test.js, authLogin.test.js, reportApi.test.js, security.test.js, loyaltyReadApi.test.js, inventory.imageUpload.test.js, doctorSession.api.test.js, and reorderThresholdApi.test.js. These suite-level results strengthen the integration evidence. They are not a complete method-and-path assertion audit for all 73 inventoried routes.

Outstanding acceptance boundaries

- Critical business logic: overall branches are below 90%, and previously identified low-coverage frontend modules retain their measured gaps. Backend individual critical module compliance cannot be certified without its per-file coverage artifact and an agreed list of critical modules.

- Hosted PayHere payment completion remains BLOCKED / NOT VERIFIED. Simulated provider callbacks and passing unit/integration tests are not equivalent to demonstrated hosted end-to-end payment.
- DDP-084/085/086/092 evidence must be judged from corresponding actual reports, environments and test cases. This appendix adds local Jest/Vitest results; it does not upgrade those independent outcomes to PASS without execution evidence.
- Release parity: these commits are verified as local observations before the fresh runs, not as the final deployed production revisions. Final approval requires reviewer acceptance of the documented boundary or independent deployment/CI evidence.

Sign-off assessment: local automated regression PASS (3,416/3,416). CI/release/hosted payment and 90% per-critical-module verification remain pending as described; no unsupported 100% approval claim is made.

DDP-087 | Execution provenance verification

Addendum - 26 October 2026 (Asia/Colombo, UTC+05:30)

This addendum supersedes earlier statements that the evidence-file contents were unavailable or that commit-to-test provenance was still awaiting inspection.

BACKEND - independently supplied pre-run evidence file

SHA: 47eb042a7627d4ae1f1f67b9c48c370379faa7c5

Timestamp: 2026-10-26 01:05:41 +05:30

Jest run: 118/118 suites; 2,075/2,075 tests passed; duration 346.634 s.

FRONTEND - independently supplied pre-run evidence file

SHA: 3e6fa9b5b277d9663982cb6e36a5af8c8187c97c

Timestamp: 2026-10-26 01:07:06 +05:30

Vitest run started 01:07:09; 110/110 files; 1,341/1,341 tests passed; duration 77.81 s. The reported start follows the evidence timestamp by 3 s.

VERIFICATION ASSESSMENT

SUPPORTED: Local test success and recorded before-run revision/timestamp for both repositories; the logs and evidence files together support the documented local execution baseline. Combined: 3,416 tests passed.

NOT ESTABLISHED: Exact CI job/artifact URL, deployment-release parity, full endpoint-by-endpoint integration verification, 90% branch coverage for every designated critical module, or hosted PayHere E2E success.

DDP-084/085/086/092 must retain their own independently evidenced statuses. **Hosted PayHere remains BLOCKED / NOT VERIFIED.**

Source basis: user-supplied ddp087-*-evidence.txt contents and complete local Jest/Vitest console logs provided for the 26 October 2026 runs.